

★ WEEK 4 · PART 2 · PRACTICE

Seasonality and SARIMA, Worked by Hand

Difference Away the Calendar, Then Decode the Six Orders

Seasonal differencing, the airline model, and combining d with D

Part 2's tools, with the arithmetic shown. We seasonally difference a quarterly series so the repeating pattern cancels, decode the six-integer SARIMA notation and the famous airline model, then combine a regular and a seasonal difference to make a trending, seasonal series stationary.

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PART 2 · APPLICATIONS

Two Timescales, Two Differences

Case A: seasonally difference a quarterly series - subtract the value from one season ago - and watch the repeating pattern vanish. Case B: decode SARIMA and the airline model, then combine a seasonal and a regular difference to reach stationarity. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

Seasonality repeats on a fixed cycle

A pattern that recurs every s periods - a quarter, a month, a week. Plain ARIMA's near-lag memory can't reach it.

2

Seasonal differencing subtracts a season ago

Replace each value with its change from s periods back. The repeating pattern cancels, leaving the underlying movement.

3

SARIMA has two sets of orders

Six integers: regular ones for short-run dynamics and trend, seasonal ones for the cycle at lag s . The airline model is the classic.

Everything here is a calculator exercise: one seasonal difference, one decoding, one combined difference.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a seasonal series.
- Seasonally difference, or combine two differences.
- Then read what's left.

What you should watch

- How subtracting a season ago cancels the cycle.
- How the six SARIMA orders split the work.
- How d and D together reach stationarity.

Seasonal Differencing

Subtraction is seasonally adjusted

A quarterly series peaks every fourth quarter - a strong, repeating season a plain difference can't remove. Seasonal differencing does: subtract the value from exactly one season, four quarters, ago. If the season repeats, those subtractions cancel it, leaving only the year-over-year change. We compute it and watch the pattern

The setup

- Two years of quarterly data.
- A Q4 peak and Q3 dip each year.
- Period s is four.

What we'll do by hand

- Subtract each quarter from the same quarter last year.
- See the seasonal swing cancel.
- Read the leftover growth.

Seasonal Differencing

One season back, not one step

$$\Delta_s y_t = y_t - y_{t-s} \quad (\text{subtract the value from one season ago; here } s=4 \text{ quarters})$$

Why not a plain difference

- A regular difference reaches only one step back.
- It can't see the pattern s periods away.
- The season would survive it.

What seasonal differencing does

- It reaches back a full season.
- Same quarter, same month, last cycle.
- The repeating pattern cancels.

Subtract Last Year's Quarter

Same quarter, one year back

	Q1	Q2	Q3	Q4
Year 1	100	110	105	140
Year 2	108	118	113	148
Delta-4	+8	+8	+8	+8

Subtracting each quarter from the same quarter a year earlier gives +8 every time. The Q4 peak minus last year's Q4 peak, the Q3 dip minus last year's dip - the season cancels exactly, leaving only the growth.

The Season Cancels

A constant is what's left

$$\Delta_4 y = 108 - 100, 118 - 110, 113 - 105, 148 - 140 = 8, 8, 8, 8 \text{ (the season cancels)}$$

What happened

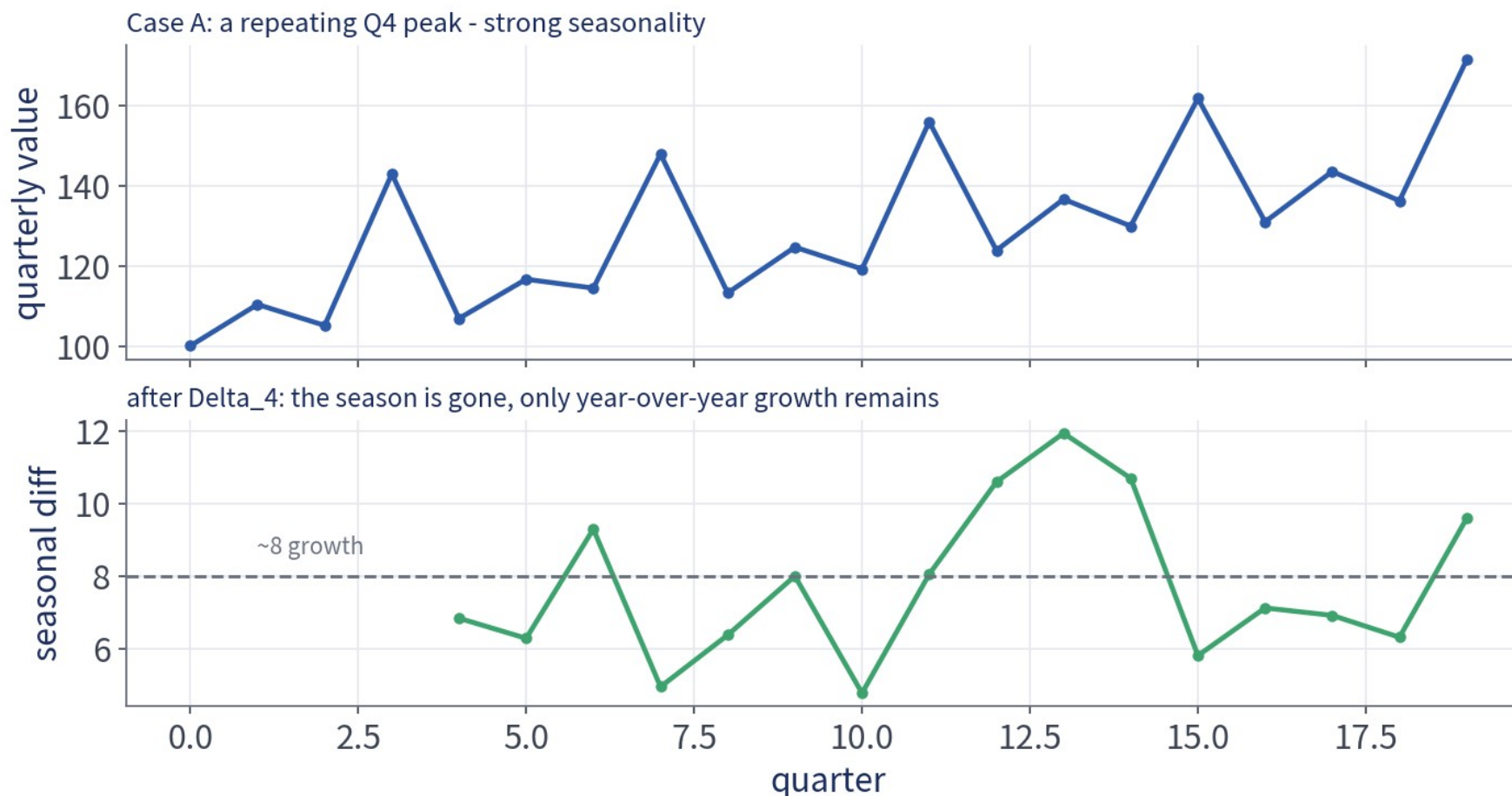
- Every seasonal difference is the same 8.
- The repeating swing has vanished.
- Only the year-over-year growth remains.

What it means

- The seasonal difference is stationary here.
- So the seasonal order D is one.
- No leftover trend, so no regular difference needed.

The Season Disappears

The top panel shows the raw series, spiking every fourth quarter - an unmistakable seasonal pattern. The bottom panel shows the seasonal difference, which flattens to a near-constant around the year-over-year growth. Subtracting one season ago has erased the calendar pattern entirely.



Additive or Multiplicative

Which season, which difference

Additive seasonality

- The swings stay a constant size.
- A seasonal difference removes them cleanly.
- That's the case we just worked.

Multiplicative seasonality

- The swings grow with the level.
- Take logs first, then difference.
- Logs turn the growing swing into a constant one.

The Airline Model

Six orders and two differences combined

When a series has both a trend and a season, one difference isn't enough. SARIMA carries six orders - regular ones for the short run and trend, seasonal ones for the cycle. The airline model is the classic. We decode it, then combine a seasonal and a regular difference on a series that has both, watching each difference

The two jobs

- The seasonal difference removes the cycle.
- The regular difference removes the trend.
- Together they reach stationarity.

What we'll do by hand

- Decode the airline model's six orders.
- Seasonally difference a trending series.
- Then difference again to flatten it.

SARIMA, Decoded

Two sets of orders, one at each scale

$\text{SARIMA}(p, d, q)(P, D, Q)_s$: (p, d, q) short-run & trend, $(P, D, Q)_s$ the season at lag s

The regular orders

- Lower-case p, d, q .
- Short-run autoregression, differencing, and MA.
- Exactly the ARIMA you already know.

The seasonal orders

- Upper-case P, D, Q at lag s .
- The same three ideas, one season apart.
- They handle the calendar cycle.

The Airline Model

The most famous SARIMA

airline model: $\text{SARIMA}(0, 1, 1)(0, 1, 1)_{12}$ – $d=1$ trend, $D=1$ season, $q=Q=1$ MA terms

Reading the orders

- d is one: a regular difference for the trend.
- D is one: a seasonal difference for the cycle.
- q and Q are one: an MA at each scale.

Why it endures

- Two differences and two MA terms.
- Fits a huge range of seasonal series.
- Named for the airline-passenger data it modeled.

Combining d and D

Season first, then trend

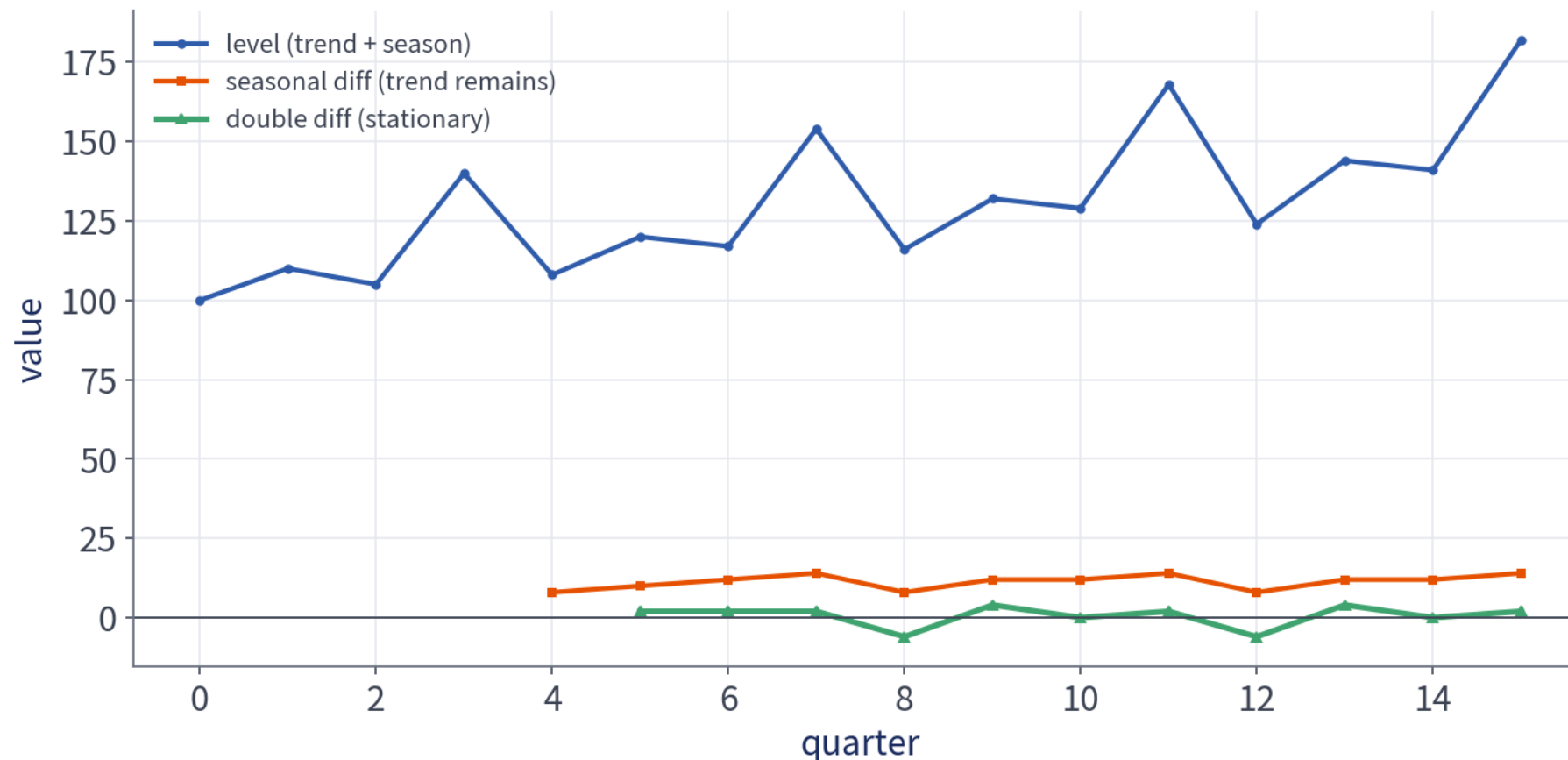
$$\Delta_4 y = 8, 10, 12, 14 \xrightarrow{\Delta} 2, 2, 2 \quad (D=1 \text{ removes season, } d=1 \text{ removes leftover trend})$$

On a series with both, the seasonal difference gives 8, 10, 12, 14 - the season is gone, but a trend remains. A regular difference of that yields a constant 2. Both differences together - D and d each one - make it stationary.

Two Differences to Flatness

The blue level carries both a trend and a season. The orange seasonal difference removes the cycle but still slopes upward. The green double difference - seasonal then regular - lands flat around a constant: stationary at last. Each difference does one job and both are needed.

Case B: seasonal then regular differencing -> a flat, stationary series



Seasonality in Finance

Powerful, but treacherous for returns

Where it helps

- Volume, earnings, and macro series cycle.
- SARIMA models those calendar effects well.
- Forecasts respect the known season.

Where it misleads

- Return seasonality is mostly noise or arbitrated away.
- Fitting a season to returns overfits.
- Use it for flows and fundamentals, not prices.

The SARIMA Workflow

Seasonal trend model for trend.

The routine extends ARIMA to two timescales. Seasonally difference to remove the calendar cycle, and regularly difference to remove the trend - using tests to fix D and d . Model what remains as a seasonal ARMA, identifying the regular and seasonal orders. Then forecast and integrate both differences back. The airline model is

The season (Case A)

- Delta-4 gave a constant 8.
- The Q4 peak cancelled exactly.
- Seasonal order D is one.

The combination (Case B)

- Seasonal diff 8,10,12,14 - trend remains.
- Regular diff of that: constant 2.
- Airline model: d and D both one.

Reproduce the Differences and the Model

Direct Claude Code to check your arithmetic

GOAL Seasonally difference, combine with a regular difference, and fit a SARIMA.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - seasonal difference:  
* for quarterly [100,110,105,140,108,118,113,148],  
  print the seasonal difference (lag 4).  
CASE B - combine and fit:  
* for [100,110,105,140,108,120,117,154], print the  
  seasonal diff, then the regular diff of that;  
* fit SARIMAX with order (0,1,1) and seasonal  
  (0,1,1,4) and print the summary.  
Then plot the level, seasonal diff, and double diff.
```

Check against the slides

- Case A seasonal diff: all 8.
- Case B seasonal diff: 8,10,12,14.
- then regular diff: 2,2,2.
- airline: (0,1,1)(0,1,1).

Deliverable: part2_practice.py whose printed numbers match the seasonal and combined differences.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a seasonal difference subtract?

The value from one full season ago

Q2 What are the six SARIMA orders?

Regular p, d, q and seasonal P, D, Q at lag s

Q3 When do you need both d and D ?

When the series has both a trend and a season

All three come from the seasonal difference, the SARIMA notation, and the combined differencing.

Three Pitfalls to Avoid

Where SARIMA trips people

Common mistakes

- "A regular difference removes the season too."
- "Difference seasonally as many times as it takes."
- "Fit a season to stock returns."

The reality

- Only a seasonal difference reaches the cycle at lag s .
- Usually D of one suffices - over-differencing adds noise.
- Return seasonality is mostly noise - use flows instead.

Two Calculations to Remember

Part 2 applications in one screen

1

Seasonal differencing cancels the cycle

Subtracting each quarter from the same quarter a year earlier gave a constant +8 - the Q4 peak and Q3 dip cancelled exactly. One seasonal difference removed the repeating pattern, leaving only year-over-year growth.

2

Combine d and D for trend plus season

On a series with both, the seasonal difference gave 8, 10, 12, 14 - season gone, trend left - and a regular difference of that gave a constant 2. The airline model, SARIMA(0,1,1)(0,1,1)₁₂, encodes exactly this: one difference at each scale.

3

Two timescales, one framework

Seasonal differencing for the calendar, regular differencing for the trend, then a seasonal ARMA on what remains. Powerful for volume, earnings, and macro series - but treacherous for returns.

Next — Part 3 practice: the full ARIMA modeling routine, from identification through diagnostics on real data.